

Ch 9. Switched Singularly Perturbed Systems with Time Delay

§ 9.1 Problem Formulation.

$$\begin{cases} \dot{x}(t) = f_{\sigma(t)}(t, x(t), x_t, z(t), z_t) & x \in \mathbb{R}^m, z \in \mathbb{R}^n, 0 < \epsilon \ll 1, \sigma: [t_0, \infty) \rightarrow \{1, 2, \dots, N\} \\ \epsilon \dot{z}(t) = g_{\sigma(t)}(t, x(t), x_t, z(t), z_t), t \geq t_0 \end{cases} \quad x(t) = x(t+s), \quad \|x_t\|_r = \sup_{t-s \leq \theta \leq t} \|x(\theta)\|$$

Assume $(x, z)^T \equiv (0^T, 0^T)$ is the trivial sol.

Def 9.1 The trivial sol of (9.2) is exponentially stable if $\exists K, \lambda > 0$ s.t.

$$\|x(t)\| + \|z(t)\| \leq K (\|x_{t_0}\|_2 + \|z_{t_0}\|_2) e^{-\lambda(t-t_0)}, \quad t \geq t_0$$

§ 9.2 Stability Analysis.

$$I_u = \{1, 2, \dots, r\}, \quad I_s = \{r+1, \dots, N\}$$

unstable mode stable mode

9.2.1 Linear Systems.

$$\begin{cases} \dot{x} = A_{11_i}(t)x + A_{12_i}(t)x_t + B_{11_i}(t)z + B_{12_i}(t)z_t \\ \epsilon \dot{z} = A_{21_i}(t)x + A_{22_i}(t)x_t + B_{21_i}(t)z \end{cases} \quad t \in [t_{k-1}, t_k) \quad \begin{matrix} * \text{ subsystem} \\ \dot{x} = A_{11_i}(t)x \\ \dot{z} = B_{21_i}(t)z \end{matrix}$$

$i \in I = I_u \cup I_s$. $A_{rs_i}, B_{rs_i}, B_{21_i}$ ($r, s=1, 2$) are conti matrices.

$A_{11_i} \in \mathbb{R}^{m \times m}, B_{11_i} \in \mathbb{R}^{m \times n}, A_{21_i} \in \mathbb{R}^{n \times m}, B_{21_i} \in \mathbb{R}^{n \times n}$

Thm 9.1 ~~(9.3)~~ The trivial sol of (9.3) is globally exponentially stable if the following hold:

- (A1) $\exists \alpha, \mu > 0$ s.t.
- (i) $\forall i \in I_u, \operatorname{Re}[\lambda(A_{11_i}(t))] > \alpha, \operatorname{Re}[\lambda(B_{21_i}(t))] \leq -\alpha$
 - (ii) $\forall i \in I_s, \operatorname{Re}[\lambda(A_{11_i}(t))] \leq -\alpha, \operatorname{Re}[\lambda(B_{21_i}(t))] \leq -\alpha$
 - (iii) $\forall i \in I, t \in [t_{k-1}, t_k) (k \in \mathbb{N})$.

$$\max \{ \|A_{11_i}(t)\|, \|\dot{A}_{11_i}(t)\|, \|B_{21_i}(t)\|, \|\dot{B}_{21_i}(t)\|, \|B_{21_i}^{-1}(t)A_{21_i}(t)\|, \|B_{21_i}^{-1}(t)A_{12_i}(t)\| \} \leq \beta$$

(A2) $P_1(t) := -B_{21_i}^{-1}(t)[A_{21_i}(t)x + A_{22_i}(t)x_t]$, $P_1(t), P_2(t)$: solutions of Lyapunov matrix eqs.

$$\begin{cases} A_{11_i}^T(t)P_1(t) + P_1(t)A_{11_i}(t) = -I_m \\ B_{21_i}^T(t)P_2(t) + P_2(t)B_{21_i}(t) = -I_n \end{cases} \quad \text{identity matrix}$$

Assume $\exists$ bdd ftns $a_{rs_i}(t), b_{rs_i}(t)$ ($r, s=1, 2$) satisfying

$$2x^T P_{1_i}(t)[A_{12_i}(t)x_t + B_{11_i}(t)z + B_{12_i}(t)z_t] + x^T \dot{P}_{1_i}(t)x \leq$$

$$a_{11_i}(t)\|x\|^2 + a_{12_i}(t)\|x_t\|_2^2 + b_{11_i}(t)\|z - h_i\|^2 + b_{12_i}(t)\|(z - h_i)_t\|_2^2,$$

$$-2(z - h_i)^T P_{2_i}(t)h_i + (z - h_i)^T \dot{P}_{2_i}(t)(z - h_i) \leq$$

$$a_{21_i}(t)\|x\|^2 + a_{22_i}(t)\|x_t\|_2^2 + b_{21_i}(t)\|(z - h_i)\|^2 + b_{22_i}(t)\|(z - h_i)_t\|_2^2.$$

(A3) (i) $\forall i \in I_u$, $d(t) := \lambda(\tilde{A}_i^T(t) + \tilde{A}_i(t))$, $\|\tilde{B}_i(t)\| \leq \beta_i$ where

$$\tilde{A}_i(t) = \begin{pmatrix} \frac{2r + a_{11,i}(t)}{\lambda_{1u}} & \frac{b_{11,i}(t)}{\lambda_{2u}} \\ \frac{a_{21,i}(t)}{\lambda_{1u}} & -\frac{1 - \varepsilon_i b_{21,i}(t)}{\varepsilon_i \lambda_{2u}} \end{pmatrix}, \quad \tilde{B}_i(t) = \begin{pmatrix} \frac{a_{12,i}(t)}{\lambda_{1u}} & \frac{b_{12,i}(t)}{\lambda_{2u}} \\ \frac{a_{22,i}(t)}{\lambda_{1u}} & \frac{b_{22,i}(t)}{\lambda_{2u}} \end{pmatrix}$$

$r > 0$ s.t. $A_{11,i} - rI$ has eigenvalues with negative real parts.

Assume $d(t) + \|\tilde{B}_i(t)\| \leq \beta_2$ ($\beta_2 > 0$)

(ii) $\forall i \in I_s$, $\exists \varepsilon_i^*, \eta > 0$ s.t. $-\tilde{A}_i(t)$ is an M -matrix, $\lambda(\tilde{A}_i(t) + \tilde{A}_i^T(t)) + 2\|\tilde{B}_i(t)\| \leq -\eta < 0$,

where

$$\tilde{A}_i(t) = \begin{pmatrix} -\frac{1 - a_{11,i}(t)}{\lambda_{1u}} & \frac{b_{11,i}(t)}{\lambda_{2u}} \\ \frac{a_{21,i}(t)}{\lambda_{1u}} & -\frac{1 - \varepsilon_i^* b_{21,i}(t)}{\varepsilon_i^* \lambda_{2u}} \end{pmatrix}, \quad \tilde{B}_i(t) = \begin{pmatrix} \frac{a_{12,i}(t)}{\lambda_{1u}} & \frac{b_{12,i}(t)}{\lambda_{2u}} \\ \frac{a_{22,i}(t)}{\lambda_{1u}} & \frac{b_{22,i}(t)}{\lambda_{2u}} \end{pmatrix}$$

$$\lambda_{1u} = \min \{ \lambda_{\min}(P_{r_i}), i \in S \}, \quad \lambda_{1u} = \max \{ \lambda_{\max}(P_{r_i}), i \in S \}, \quad r = 1, 2$$

* M -matrix: off-diagonal entries are less than or equal to zero
+ eigenvalues have nonnegative real parts.

(A4) Let $\lambda^+ = \max \{ \xi_i \mid \forall i \in I_u \}$, $\lambda^- = \min \{ \xi_i \mid \forall i \in I_s \}$

$T^+(t_0, t)$, $T^-(t_0, t)$: total activation times
↑ unstable ↑ stable

Assume that $\forall t_0 \in \mathbb{R}_+$, $\inf_{t \geq t_0} \frac{T^-(t_0, t)}{T^+(t_0, t)} \geq \frac{\lambda^+ + \lambda^*}{\lambda^- - \lambda^*}$ where $\lambda^* \in (0, \lambda^-)$.

Furthermore, $\exists 0 < \nu < \lambda^*$ s.t.

(i) $i \in I_u \Rightarrow \ln \mu - \nu(t_k - t_{k-1}) \leq 0, \quad k = 1, 2, \dots, r$

(ii) $i \in I_s \Rightarrow \ln \mu + \xi_i(t - t_{k-1}) \leq 0 \quad k = r+1, \dots, N$

where ξ_i is a unique positive sol of

$$\xi_i = \lambda(\tilde{A}_i^T + \tilde{A}_i) + \|\tilde{B}_i\| + \|\tilde{B}_i\| e^{\xi_i t}$$

$$\mu \leq e^{\nu(t_k - t_{k-1})}$$

$$\mu \leq e^{\frac{-\xi_i(t)}{e^{-\xi_i t}}} e^{\nu(t_k - t_{k-1})}$$

(pf) For $i \in I$, $V_i(x(t)) := x^T(t) P_i(t) x(t)$, $W_i((z-h_i)(t)) := (z-h_i)^T(t) P_{2i}(t) (z-h_i)(t)$ as the Lyapunov ftn.

For $i \in I_u$, we have

$$\begin{aligned} \dot{V}_i(x(t)) &= \dot{x}^T P_i(t) x + x^T \dot{P}_i(t) x + x^T P_i(t) \dot{x} \\ &\leq \left(2\gamma + \frac{a_{11i}(t)}{\lambda_{1m}}\right) V_i(t) + \frac{b_{11i}(t)}{\lambda_{2m}} W_i(t) + \frac{a_{12i}(t)}{\lambda_{1m}} \|V_i\|_Z \\ &\quad + \frac{b_{12i}(t)}{\lambda_{2m}} \|W_i\|_Z \end{aligned} \quad \text{where } W_i(t) := W_i((z-h_i)(t))$$

$$\begin{aligned} W_i((z-h_i)(t)) &\leq \frac{a_{21i}(t)}{\lambda_{1m}} V_i(t) - \frac{1-\varepsilon_i b_{21i}(t)}{\varepsilon_i \lambda_{2m}} W_i(t) + \frac{a_{22i}(t)}{\lambda_{1m}} \|V_i\|_Z \\ &\quad + \frac{b_{22i}(t)}{\lambda_{2m}} \|W_i\|_Z \\ &\leq \frac{a_{21i}(t)}{\lambda_{1m}} V_i(t) - \frac{1-\varepsilon_i^* b_{21i}(t)}{\varepsilon_i^* \lambda_{2m}} W_i(t) + \frac{a_{22i}(t)}{\lambda_{1m}} \|V_i\|_Z + \frac{b_{22i}(t)}{\lambda_{2m}} \|W_i\|_Z \\ &\quad (\varepsilon_i^* \geq \varepsilon_i > 0) \end{aligned}$$

$$\Rightarrow \begin{pmatrix} \dot{V}(t) \\ \dot{W}(t) \end{pmatrix} \leq \begin{pmatrix} \frac{2\gamma + a_{11i}(t)}{\lambda_{1m}} & \frac{b_{11i}(t)}{\lambda_{2m}} \\ \frac{a_{21i}(t)}{\lambda_{1m}} & -\frac{1-\varepsilon_i b_{21i}(t)}{\varepsilon_i \lambda_{2m}} \end{pmatrix} \begin{pmatrix} V(t) \\ W(t) \end{pmatrix} + \begin{pmatrix} \frac{a_{12i}(t)}{\lambda_{1m}} & \frac{b_{12i}(t)}{\lambda_{2m}} \\ \frac{a_{22i}(t)}{\lambda_{1m}} & \frac{b_{22i}(t)}{\lambda_{2m}} \end{pmatrix} \begin{pmatrix} \|V_i\|_Z \\ \|W_i\|_Z \end{pmatrix}$$

by A3(i) & Lemma 2.4, $\exists \xi_i$ s.t. (pg 15)

$$\begin{aligned} V_i(x(t)) &\leq (\|V_{i,t_{k-1}}\|_Z + \|W_{i,t_{k-1}}\|_Z) e^{\xi_i(t-t_{k-1})} \\ W_i((z-h_i)(t)) &\leq (\|V_{i,t_{k-1}}\|_Z + \|W_{i,t_{k-1}}\|_Z) e^{\xi_i(t-t_{k-1})} \end{aligned}$$

For $i \in I_s$, $\dot{V}_i(t) \leq \left(\frac{2\gamma + a_{11i}(t)}{\lambda_{1m}} - \frac{1-a_{11i}(t)}{\lambda_{1m}} - \frac{b_{11i}(t)}{\lambda_{2m}} \right) V_i(t) + \left(\frac{a_{12i}(t)}{\lambda_{1m}} - \frac{1-\varepsilon_i^* b_{21i}(t)}{\varepsilon_i^* \lambda_{2m}} \right) W_i(t) + \left(\frac{a_{22i}(t)}{\lambda_{1m}} - \frac{b_{22i}(t)}{\lambda_{2m}} \right) (\|V_i\|_Z + \|W_i\|_Z)$

by A3(ii) & Lemma 2.3, $\exists \xi_i$ s.t. (pg 14)

$$V_i(t) \leq (\|V_{i,t_{k-1}}\|_Z + \|W_{i,t_{k-1}}\|_Z) e^{-\xi_i(t-t_{k-1})}, \quad W_i(t) \leq (\|V_{i,t_{k-1}}\|_Z + \|W_{i,t_{k-1}}\|_Z) e^{-\xi_i(t-t_{k-1})}$$

$\forall i, j \in I$, $V_i(t) \leq \mu V_j(t)$, $W_j(t) \leq \mu W_i(t)$, $\mu = \max \left\{ \frac{\lambda_{2m}}{\lambda_{1m}}, \frac{\lambda_{1m}}{\lambda_{2m}} \right\}$, $\ell = 1, 2$

$$\Rightarrow V_N(x(t)) \leq \prod_{i=1}^r 2\mu e^{\xi_i(t_i-t_{i-1})} \times \prod_{j=r+1}^{N+1} 2\mu e^{\xi_j t_j} e^{-\xi_j(t_j-t_{j-1})} \times (\|V_{1,t_0}\|_Z + \|W_{1,t_0}\|_Z) e^{-\xi_N(t-t_{N-1})}$$

by A4, $V_N(t) \leq (\|V_{1,t_0}\|_Z + \|W_{1,t_0}\|_Z) e^{-(\theta^*-2)(t-t_0)}$

Similarly $W_N((z-h_N)(t)) \leq (\|V_{1,t_0}\|_Z + \|W_{1,t_0}\|_Z) e^{-(\lambda^*-2)(t-t_0)}$

$\exists K_1$ s.t.

$$\Rightarrow \|x(t)\| \leq K_1 (\|x_{t_0}\|_Z + \|z_{t_0}\|_Z) e^{-(\lambda^* - \nu)(t - t_0)/2}$$

$$\|z\| - \|h_i\| \leq \|z - h_i\| \leq \frac{1}{\sqrt{\lambda_{\min}}} W_i^{1/2}$$

$$\exists K_2 \text{ s.t. } \|z(t)\| \leq K_2 (\|x_{t_0}\|_Z + \|z_{t_0}\|_Z) e^{-(\lambda^* - \nu)(t - t_0)/2}$$

$$\Rightarrow \|x(t)\| + \|z(t)\| \leq K (\|x_{t_0}\|_Z + \|z_{t_0}\|_Z) e^{-(\lambda^* - \nu)(t - t_0)/2}, \quad K = K_1 + K_2$$

□

Remark $A_1(i)(iii)$, A_2 , $A_3(i)$: $I_u \ni i$ is unstable

$A_1(ii)(iii)$, $A_3(ii)$: $i \in I_s$ exp stable

(A4) : total activation time of stable subsystem is larger than that of unstable ones

Ex 9.1

$$\begin{cases} \dot{x} = 0.1x + 0.04z(t-1) \\ \dot{z} = 0.45x(t-1) - 0.1z \end{cases}$$

unstable

$$\begin{cases} \dot{x} = -1.5x - 0.1z(t-1) \\ \dot{z} = -x(t-1) - 2z \end{cases}$$

stable

$\Rightarrow$ Fig 9.1

§9.22. Nonlinear systems

Thm 9.2

almost same

Ex 9.2

$$\begin{cases} \dot{x} = 0.1x + \sin z(t-1) \\ \dot{z} = 0.1x - z \end{cases} \quad \text{unstable}$$

$$\begin{cases} \dot{x} = -10x + \ln(1+x^2(t-1)) + z \\ \dot{z} = x - 2z \end{cases} \quad \text{stable}$$